

Project Design Phase
Problem – Solution Fit Template

Date	14 JUNE 2026
Team ID	INDIVIDUAL PROJECT
Project Name	ShopEZ: E-commerce Application
Maximum Marks	2 Marks

Problem – Solution Fit Template:

The target customer for ShopEZ consists of frequent, tech-savvy mobile shoppers who operate in a fast-paced environment with highly limited time. Currently, these users face significant pain points when attempting to make purchases, such as broken or unhelpful search filters, clunky and long checkout sequences, and lagging performance on mobile browsers. These functional barriers are accompanied by emotional ones, specifically anxiety and nervousness regarding weak security systems during data submission. Looking deep into the system architectures, the root causes of these issues rest entirely on unoptimized database queries, a lack of modern authentication tokens, and poor mobile-first layout designs.

Designing the ShopEZ Solution and Adoption Strategy

To bridge the gap between user frustrations and system architecture, the ShopEZ platform introduces targeted technical features designed to perfectly match the customer's exact needs. The platform integrates optimized search indices to allow fast findings, alongside a simplified, one-step checkout workflow. Security concerns are neutralized by deploying modern JSON Web Token (JWT) user authentication protocols within a fully responsive MERN stack frontend architecture. To drive adoption seamlessly, the strategy targets these fast-paced individuals across mobile channels, leveraging the app's snappy performance indicators and prominent security badges to build swift user trust.

Communication Channels and Behaviour Triggers

Understanding the behavioural patterns of the target demographic is crucial for shaping the platform's positioning. Mobile users naturally display an existing tendency to shop during quick work breaks or daily commutes, relying heavily on peer reviews for digital trust, and will instantly exit any e-commerce application that displays lag. ShopEZ taps directly into this behavioral medium by focusing all user communication on themes of extreme time-savings and bulletproof security. By highlighting specific speed metrics and responsive mobile app storefront interactions in marketing materials, the platform triggers immediate user engagement from shoppers seeking a reliable alternative.

Eliminating Annoyances and Urgent Problems

Ultimately, achieving a true Problem-Solution fit relies on resolving immediate user irritations while securing high-impact outcomes. ShopEZ systematically eliminates frequent daily annoyances by fixing broken checkout validation errors and eradicating loading lag from product search grids. On a critical level, the system resolves the most urgent and costly business problems by safeguarding vital customer information against vulnerabilities and preventing cart abandonment. By converting a historically slow, unoptimized experience into a secure, instant transactional pipeline, ShopEZ successfully matches the behavioural habits of the modern mobile consumer.

Template:

ROW 1: TOP PANEL – FOCUS: DEFINE CS, FIT INTO CC EXPLORE AS, DIFFERENTIATE		
1 CUSTOMER SEGMENT(S) [CS] Target: Everyday mobile online shoppers. Profile: Tech-savvy but time-poor individuals who balance busy routines, daily commutes, and quick work breaks. Habits: Highly dependent on smartphones for immediate retail needs and expect immediate app responsiveness.	6 CUSTOMER CONSTRAINTS [CC] Time limits: Needs to make purchases in short, fragmented bursts of free time. Network shifts: Shifting mobile data speeds (LTE/5G) while traveling on the go. Trust barrier: Strictly refuses to enter credit card details into slow or insecure-looking interfaces.	5 AVAILABLE SOLUTIONS [AS] Alternatives: Heavy desktop e-commerce websites or slow, legacy web storefronts. Pros: Offer large product catalogs. Cons: Poor mobile responsive layouts, massive loading lag, and frustrating multi-page checkout forms.
ROW 2: MIDDLE PANEL – FOCUS: FOCUS ON J&P, TAP INTO BE, UNDERSTAND RC		
2 JOBS-TO-BE-DONE / PROBLEMS [J&P] Core Job: Find products effortlessly, add to a smooth cart, and check out securely on a phone. Pain Points: Interface is laggy, search filters return empty or broken results, and the transaction screen hangs.	7 BEHAVIOUR [BE] Action: Compares multiple retail options on mobile tabs simultaneously. Reaction: Instantly leaves the store and completes a cart abandonment if a page takes more than a few seconds to load. Pattern: Actively looks for secure checkout indicators before buying.	9 PROBLEM ROOT CAUSE [RC] Backend: The application database lacks proper indexing, creating severe lag during product search queries. Security: Missing stateless token guards (JWT), making the session management clunky and vulnerable. Frontend: The UI lacks mobile-first responsiveness.
ROW 3: BOTTOM PANEL – FOCUS: IDENTIFY STRONG TR & EM EXTRACT ONLINE & OFFLINE CH OF BE		
3 TRIGGERS [TR] Internal: An urgent need to buy an item quickly during a short break. External: High irritation with a competing app that just crashed or lagged, or seeing a peer use a fast shopping app.	10 YOUR SOLUTION [SL] – SHOPEZ Architecture: A snappy e-commerce web application built on the MERN Stack. Key Pillars: ✓ Optimized database search indexing for instant results. ✓ One-step checkout guarded by secure JSON Web Tokens (JWT). ✓ A fully fluid, responsive design tailored for mobile viewports.	8 CHANNELS OF BEHAVIOUR [CH] 8.1 ONLINE: - Uses search keywords on mobile browsers, follows trusted links from social media platforms, and reviews tech security badges. 8.2 OFFLINE: - Shops while multitasking during train commutes, asks colleagues face-to-face for app recommendations, or leaves digital stores entirely to run physical errands when apps fail.